PS 03: Forecasting Liquid & Overnight net flows and sizing the cash buffer

All numbers below are produced by `python -m liqfc run` and rendered from `outputs/*.csv` by `scripts/report_tables.py` and `scripts/build_report.py` (this file is generated from `reports/report_template.md`; the numbers in sentences are filled in from the CSVs). Units are ₹ crore; negative net flow = net outflow.

> **Penalty definition (read this first).** The brief prices decision cost as "yield given up on idle cash + a 3× penalty on any redemption shortfall". This report uses **reading (a), primary:** each rupee short costs **3 × the monthly idle-cash spread** (3 × shortfall × s/1200, where s is FBIL 3M T-bill − WACR, floored at 0.10%). **Reading (b)** charges **3 × the shortfall amount itself** (₹3 per ₹1 short). Both are computed everywhere (`outputs/decision_costs.csv`, column `penalty_basis`; §3a; dashboard). **Yes, the conclusion flips.** Under reading (a) q75 is cheaper than q95 in 11 of 12 category × model cases and q95 in 1; under reading (b) q75 is cheaper than q95 in 0 of 12 category × model cases and q95 in 12. The cheaper policy changes in 11 of 12 cases. The forecast-based q75 buffer from `sarimax_cal` beats the naive baseline under **both** readings in both categories (Liquid -26% / -40%; Overnight -34% / -37%). The point estimates favour the forecast buffer, but under reading (a) the Liquid gain is not statistically clear: its 90% bootstrap CI of the 24-month difference [-33.3, +23.0] ₹cr includes 0 (Overnight: 90% bootstrap CI of the 24-month difference [-30.1, -11.3] ₹cr excludes 0). The choice between q75 and q95 depends on the reading.

1. Data

- **AMFI** monthly reports (official, `amfiindia.com/spages/am{mon}{yyyy}repo.xls`). There are 102 genuine workbooks. 75 are xlsx content saved under a .xls name and 27 are OLE2 xls, so the parser picks the engine from magic bytes. 42 URLs (Jan-2015 to Mar-2018 and Oct-Dec-2026) returned 404 HTML pages. Those pages were not kept, and the loader rejects any non-workbook and logs it.
  - Apr-2018 to Mar-2019 files use the old sponsor-wise layout, with a single combined Liquid/Money-Market row, so they are excluded. **The panel runs 2019-04 to 2026-09: 90 months × 2 categories.**
  - Validation passes: no duplicates or gaps, sales − redemptions = net within 0.1 cr (max residual ~1e-9), no negative gross flows, AUM-continuity unit checks hold, and the panel matches the auditor's independent extract on all 180 rows.
  - Six 2020 files have no unit text. Their units were confirmed through AUM continuity.
  - One robust outlier is flagged and kept: Overnight 2019-11.
  - AMFI figures are first releases and are treated as provisional.
- **Release dates** come from the "Released on" stamp: the lag after month-end is 8 to 13 days. Where a file has no stamp (2019-05, 2020-01, 2021-12 and 2025-09 to 2026-09) the date is set to month-end + 13 days, the maximum observed lag.
- **Drivers** (`data/raw/MANIFEST.csv` has URL, retrieval time in UTC and IST, sha256 and notes for 330 files):
  - WACR: RBI daily call rates.
  - Net LAF liquidity: RBI Money Market Operations "net liquidity from today's operations", from Jul-2020 onward.
  - FBIL 3M T-bill benchmark: official public archive.
  - Nifty 50 month-end close: NSE official archive files.
  - Daily RBI series up to 17-Sep-2026 come from `dbie.rbihub.in`. This is a **third-party DBIE mirror, not confirmed official**. It matched the official RBI press release of 15-Sep-2026 exactly (test `test_mirror_matches_official_release_sep15`). Days after 17-Sep come from official RBI press releases. 23, 26–28 and 30 Sep 2026 were not found, so September liquidity averages 23 days.
  - DBIE mirror tables that were unusable (policy rates with empty values, T-bill auctions with corrupted decimals, a Nifty table with duplicate months) are listed in the manifest and not used.

2. Method

- **Classical models:**
  - `snaive`: y(t−12), with PIs from in-sample seasonal differences.
  - `snaive_ratio`: the same, applied to flow/lagged AUM and rescaled by AUM_T.
  - `ets_ratio`: ETS(A,N,A) on that ratio.
  - `sarimax_cal`: AR(1) on the ratio with calendar dummies (Jun/Sep/Dec advance-tax quarter-ends, March FY-end, April, other quarter starts).
- **ML model:** `lgbm_q`, quantile LightGBM with 9 quantiles and sorted outputs so they cannot cross. It is pooled over both categories and all horizons, fits each horizon directly, and predicts the target y(t+h)/AUM_t.
  - Features: target-month calendar, ratio lags, the same month last year (and the year before), 12-month AUM growth, WACR level and 3-month change, T-bill−WACR spread, net LAF, and Nifty 1-month and 3-month returns.
  - Hyper-parameters were fixed in `config.yaml` beforehand and not tuned.
- **Out-of-sample, horizon-specific PIs** (`*_oosq`): at each origin, the quantiles of that model's own past out-of-sample errors (scaled by AUM) for the same category and horizon, using only targets already released. A model gets these PIs once it has at least 12 such errors.
- **Rolling-origin backtest:** origins run from 2022-03 to 2026-08 and every model is refit at each origin. The headline window is the 24 targets Oct-2024 to Sep-2026 for h = 1, 2 and 3, which gives 72 forecasts per model per category.
- **Nested selection:** each category's model is chosen on pinball loss over Oct-2024 to Sep-2025 only, then reported on Oct-2025 to Sep-2026.
- **No-leakage checks** (tests):
  - Scrambling all data after the origin leaves every model's forecast unchanged.

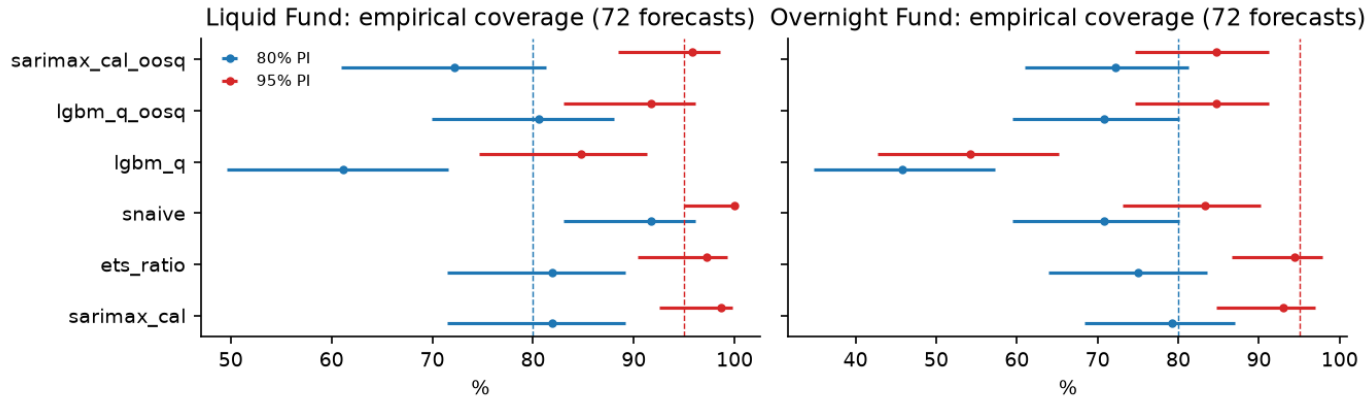
- ML features built from data truncated at T equal those built from the full data.
- A driver whose `available_date` falls after the release date is dropped.
- Backtest rows always have target > origin and as_of < target release.
- Removing the truncation in `info_set` makes 6 of these tests fail (checked by deliberately breaking the code).

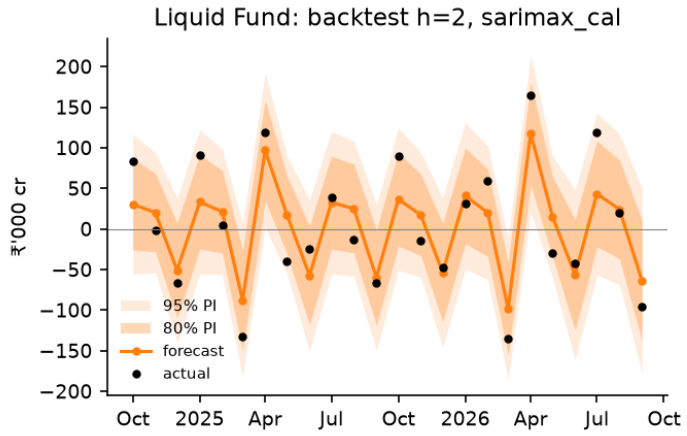
3. Results: accuracy, calibration and decision cost (24 targets × h = 1–3; decision uses h=2)

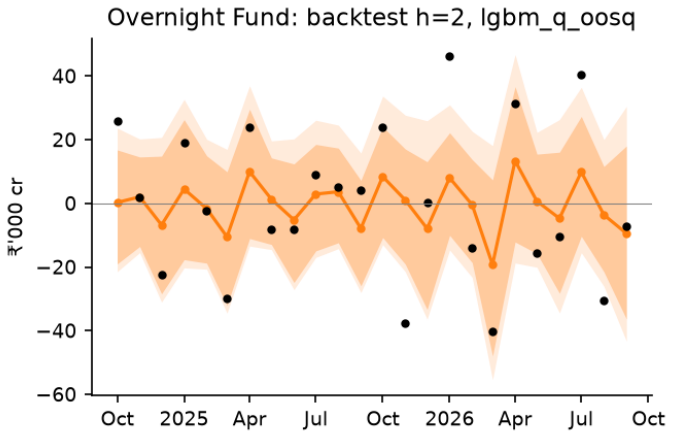
- **MAPE handling:** MAPE is reported on all points, as the spec requires. Net flows cross zero, so MAPE is dominated by small months: Overnight snaive is 497%, but only 133% once |y| < 2,000 cr is excluded. Read median APE, MASE and pinball alongside it.
- The seasonal-naive figures reproduce the auditor's reference values exactly: Liquid MAPE 137%, median APE 53%, MAE 33,199; Overnight MAPE 497%, median APE 81%, MAE 15,804.
- Coverage is pooled over 72 overlapping forecasts. Because those forecasts are not independent, the Wilson intervals are optimistic.

Category	Model	MAPE %	MAPE excl. abs(y)<2k cr %	Median APE %	sMAPE %	MAE ₹cr	MASE	Pinball (core) ₹cr	Cov80 [Wilson]	Cov95 [Wilson]	Cost q75 (a) ₹cr (vs base)	Cost q95 (a) ₹cr (vs base)
Liquid Fund	sarimax_cal_oosq	115	76	49	81	31,242	0.78	6,561	72% [61%,81%]	96% [88%,99%]	53.3 (-21%)	85.3 (+27%)
Liquid Fund	sarimax_cal	115	76	49	81	31,242	0.78	6,563	82% [72%,89%]	99% [93%,100%]	50.0 (-26%)	114.3 (+70%)
Liquid Fund	ets_ratio	146	87	50	84	33,801	0.84	7,036	82% [72%,89%]	97% [90%,99%]	66.4 (-1%)	124.0 (+85%)
Liquid Fund	ets_ratio_oosq	146	87	50	84	33,801	0.84	7,042	78% [67%,86%]	94% [87%,98%]	69.6 (+4%)	115.7 (+72%)
Liquid Fund	snaive	137	140	53	83	33,199	0.83	7,267	92% [83%,96%]	100% [95%,100%]	56.7 (-16%)	168.0 (+150%)
Liquid Fund	snaive_oosq	137	140	53	83	33,199	0.83	7,297	88% [78%,93%]	100% [95%,100%]	58.1 (-13%)	166.7 (+148%)
Liquid Fund	snaive_ratio_oosq	154	158	49	79	33,223	0.83	7,449	89% [80%,94%]	99% [93%,100%]	76.9 (+15%)	185.0 (+176%)
Liquid Fund	snaive_ratio	154	158	49	79	33,223	0.83	7,808	96% [88%,99%]	100% [95%,100%]	78.3 (+17%)	205.2 (+206%)
Liquid Fund	lgbm_q_oosq	144	82	70	107	42,429	1.06	9,068	81% [70%,88%]	92% [83%,96%]	94.2 (+40%)	142.9 (+113%)
Liquid Fund	lgbm_q	144	82	70	107	42,429	1.06	10,307	61% [50%,72%]	85% [75%,91%]	79.0 (+18%)	134.4 (+100%)
Overnight Fund	sarimax_cal	133	81	81	128	14,047	1.02	3,172	79% [68%,87%]	93% [85%,97%]	43.1 (-34%)	51.1 (-21%)
Overnight Fund	sarimax_cal_oosq	133	81	81	128	14,047	1.02	3,228	72% [61%,81%]	85% [75%,91%]	40.8 (-37%)	47.6 (-27%)
Overnight Fund	ets_ratio	207	82	73	113	14,736	1.07	3,304	75% [64%,84%]	94% [87%,98%]	42.1 (-35%)	54.6 (-16%)
Overnight Fund	ets_ratio_oosq	207	82	73	113	14,736	1.07	3,384	76% [65%,85%]	85% [75%,91%]	41.5 (-36%)	52.8 (-19%)
Overnight Fund	snaive_oosq	497	133	81	119	15,804	1.15	3,412	72% [61%,81%]	92% [83%,96%]	40.0 (-38%)	64.1 (-1%)
Overnight Fund	snaive_ratio	493	131	80	116	15,557	1.13	3,413	79% [68%,87%]	92% [83%,96%]	38.2 (-41%)	71.0 (+10%)
Overnight Fund	lgbm_q_oosq	184	76	70	115	14,816	1.08	3,423	71% [59%,80%]	85% [75%,91%]	35.4 (-45%)	41.9 (-35%)
Overnight Fund	snaive	497	133	81	119	15,804	1.15	3,426	71% [59%,80%]	83% [73%,90%]	37.6 (-42%)	58.9 (-9%)
Overnight Fund	snaive_ratio_oosq	493	131	80	116	15,557	1.13	3,444	71% [59%,80%]	86% [76%,92%]	38.4 (-41%)	55.9 (-14%)
Overnight Fund	lgbm_q	184	76	70	115	14,816	1.08	4,834	46% [35%,57%]	54% [43%,65%]	50.9 (-21%)	48.9 (-25%)

Baseline (hold last year's same-month net outflow) 24-month cost, h=2 decision: Liquid Fund: ₹67.1 cr (8 shortfall months, mean buffer ₹29,060 cr); Overnight Fund: ₹64.9 cr (10 shortfall months, mean buffer ₹6,190 cr)







Per-horizon detail: reports/tables_by_h.md. Nested selection:

Category	Model	Pinball select window (Oct24-Sep25)	Pinball eval window (Oct25-Sep26)	MAE eval	Cov80 eval	Cov95 eval	Selected
Liquid Fund	ets_ratio_oosq	7,616	6,467	30,951	81%	100%	
Liquid Fund	ets_ratio	7,464	6,608	30,951	86%	97%	
Liquid Fund	sarimax_cal_oosq	6,418	6,704	31,747	78%	92%	
Liquid Fund	sarimax_cal	6,301	6,826	31,747	83%	97%	**yes**
Liquid Fund	snaive_ratio_oosq	8,052	6,846	28,865	89%	100%	
Liquid Fund	snaive_oosq	7,650	6,945	30,895	89%	100%	
Liquid Fund	snaive	7,539	6,995	30,895	92%	100%	
Liquid Fund	snaive_ratio	8,343	7,274	28,865	100%	100%	
Liquid Fund	lgbm_q_oosq	8,098	10,038	44,219	83%	83%	
Liquid Fund	lgbm_q	9,361	11,252	44,219	64%	86%	
Overnight Fund	snaive_ratio	3,061	3,765	17,371	75%	92%	
Overnight Fund	snaive_oosq	2,931	3,893	17,445	67%	89%	
Overnight Fund	sarimax_cal	2,446	3,898	18,197	69%	89%	
Overnight Fund	snaive_ratio_oosq	2,935	3,953	17,371	61%	81%	
Overnight Fund	snaive	2,846	4,006	17,445	58%	75%	
Overnight Fund	sarimax_cal_oosq	2,438	4,018	18,197	61%	78%	
Overnight Fund	ets_ratio	2,419	4,190	19,628	67%	92%	
Overnight Fund	ets_ratio_oosq	2,552	4,217	19,628	67%	83%	
Overnight Fund	lgbm_q_oosq	2,322	4,524	19,648	58%	78%	**yes**
Overnight Fund	lgbm_q	3,105	6,564	19,648	33%	44%	

What the results show.

1. **Accuracy.** `sarimax_cal` is best for both categories on pooled pinball and MAE.
   - Liquid: pinball 6,563 vs 7,267 for snaive (−10%), MASE 0.78.
   - Overnight: pinball 3,172 vs 3,426 (−7%).
   - Its gain over seasonal-naive is modest, and Overnight MASE is about 1.0, so Overnight is essentially unforecastable beyond seasonality.
2. **The ML challenger does not beat the classical models.** Raw `lgbm_q` is the worst model on pinball and MAE for Liquid (MAE 42,429 vs 31,242), and its PIs under-cover: Overnight 80% PI covers 46% [35%, 57%]. The quantile GBM overfits on about 200 pooled training rows. Recalibrating its intervals with past out-of-sample errors (`lgbm_q_oosq`) repairs coverage (Liquid 81%) but not accuracy.
3. **Coverage.** The classical models are roughly calibrated at 80%: `sarimax_cal` gives Liquid 82% and Overnight 79%. snaive over-covers Liquid (92% / 100%), i.e. its intervals are too wide. Overnight 95% intervals tend to under-cover (83–94%).
4. **Nested selection** picks `sarimax_cal` for Liquid. For Overnight it picks `lgbm_q_oosq`, which then ranks 9 of 10 on the evaluation window (pinball 4,524; best was `snaive_ratio` (3,765)). With only 12 selection months the choice is noisy. I report this as it came out and did not re-pick on the test window.

Decision layer, penalty reading (a) (h=2, FBIL 3M − WACR spread; 90% moving-block-bootstrap CI of the 24-month total cost difference vs the baseline):

Category	Model	Policy	Total cost diff vs baseline ₹cr	90% block-bootstrap CI	Months cheaper (of 24)
Liquid Fund	sarimax_cal	q75	-17.1	[-33.3, +23.0]	7
Liquid Fund	snaive	q75	-10.4	[-26.5, +25.3]	6
Liquid Fund	ets_ratio	q75	-0.7	[-13.5, +28.8]	7
Liquid Fund	snaive_ratio	q75	+11.1	[-2.9, +42.7]	6

Category	Model	Policy	Total cost diff vs baseline ₹cr	90% block-bootstrap CI	Months cheaper (of 24)
Liquid Fund	lgbm_q	q75	+11.9	[-13.4, +64.8]	8
Liquid Fund	lgbm_q_oosq	q75	+27.0	[+1.4, +75.8]	5
Liquid Fund	sarimax_cal	q95	+47.2	[+21.9, +79.0]	5
Liquid Fund	ets_ratio	q95	+56.8	[+26.0, +82.3]	4
Liquid Fund	lgbm_q	q95	+67.3	[+36.1, +132.2]	4
Liquid Fund	lgbm_q_oosq	q95	+75.8	[+36.3, +108.2]	5
Liquid Fund	snaive	q95	+100.9	[+73.3, +140.0]	3
Liquid Fund	snaive_ratio	q95	+138.1	[+95.6, +171.5]	1
Overnight Fund	lgbm_q_oosq	q75	-29.4	[-47.6, -14.5]	10
Overnight Fund	snaive	q75	-27.3	[-42.9, -9.2]	9
Overnight Fund	snaive_ratio	q75	-26.7	[-46.0, -9.6]	9
Overnight Fund	ets_ratio	q75	-22.7	[-31.8, -9.4]	11
Overnight Fund	sarimax_cal	q75	-21.8	[-30.1, -11.3]	14
Overnight Fund	lgbm_q	q75	-13.9	[-22.2, -1.0]	11
Overnight Fund	lgbm_q_oosq	q95	-22.9	[-44.9, -7.4]	8
Overnight Fund	lgbm_q	q95	-16.0	[-30.1, -3.3]	12
Overnight Fund	sarimax_cal	q95	-13.8	[-37.1, +4.0]	8
Overnight Fund	ets_ratio	q95	-10.3	[-32.8, +9.2]	8
Overnight Fund	snaive	q95	-6.0	[-28.2, +16.1]	8
Overnight Fund	snaive_ratio	q95	+6.2	[-16.8, +23.2]	7

- **Under reading (a), the spec's 95th-percentile buffer is too conservative for Liquid.** Across the 10 forecasters it costs +27% to +206% relative to the naive baseline, and the 90% CI excludes 0 on the costly side for 9 of 10. For Overnight, q95 changes cost by -35% to +10% (9 of 10 cheaper than baseline). Under reading (b) this conclusion reverses (§3a). Under this cost function the cost-minimising quantile is the newsvendor critical ratio 3/(3+1) = **0.75**. That value is set in `config.yaml` and was not tuned on the test window.
- **The q75 policy beats the baseline for Overnight.** With `sarimax_cal` the cost changes by -34%. Across the 10 forecasters the 24-month cost changes by -45% to -21% (10 of 10 cheaper than baseline; 90% CI excludes 0 on the cheap side for 10 of 10).
- **For Liquid, q75 + `sarimax_cal` changes cost by -26%** (backtest q75, penalty reading (a): ₹17.1 cr cheaper than baseline over 24 m (90% CI includes 0)).
- The sign of these results holds when the spread is a constant 0.50% (`reports/tables_decision_sensitivity.md`). The h=1 rows there are an oracle-timing check only: the M−1 release arrives around the 10th of M, after the day-1 decision, so h=1 is not feasible live.
- **Costs are small in ₹** (tens of crore over 24 months) because the T-bill − WACR spread sat at the 0.10% floor in 18 of 24 months; WACR traded above the 3M T-bill. Rankings are therefore driven largely by shortfall versus idle amounts, not by the spread level.
- **Net, not gross, category outflow** is used. A single fund should scale the buffer by its share of category AUM.

3a. Penalty reading: both cost tables and whether the conclusion flips

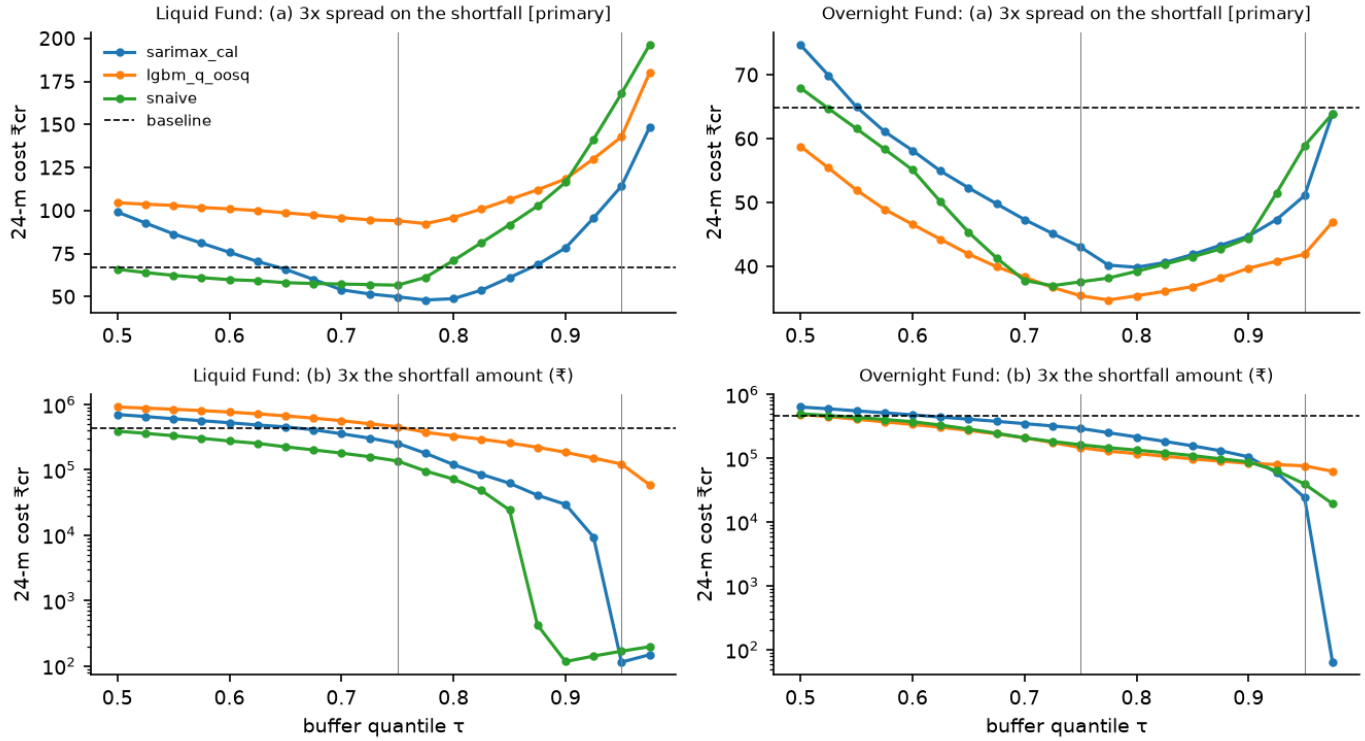
Reading (a), primary: shortfall cost = 3 × shortfall × s/1200, so idle and shortfall are priced in the same unit (₹ of carry per month) and the penalty is three times the idle rate. **Reading (b):** shortfall cost = 3 × shortfall (₹). Idle cost is idle × s/1200 under both. h=2, realised spread, 24 months Oct-2024 to Sep-2026; full tables for six models in `reports/tables_penalty.md`.

Category	Model	Policy	Breach months	Mean buffer ₹cr	(a) cost ₹cr	(a) vs baseline	(b) cost ₹cr	(b) vs baseline
Liquid Fund	baseline	smly_baseline	8	29,060	67.1	+0%	427,724	+0%
Liquid Fund	sarimax_cal	q75	6	35,399	50.0	-26%	255,233	-40%
Liquid Fund	sarimax_cal	q95	0	74,057	114.3	+70%	114	-100%
Liquid Fund	lgbm_q_oosq	q75	7	40,027	94.2	+40%	456,274	+7%
Liquid Fund	lgbm_q_oosq	q95	2	75,892	142.9	+113%	122,700	-71%
Liquid Fund	snaive	q75	4	43,702	56.7	-16%	136,712	-68%
Liquid Fund	snaive	q95	0	92,426	168.0	+150%	168	-100%
Liquid Fund	ets_ratio	q75	7	37,800	66.4	-1%	289,238	-32%
Liquid Fund	ets_ratio	q95	1	73,490	124.0	+85%	19,604	-95%
Overnight Fund	baseline	smly_baseline	10	6,190	64.9	+0%	467,918	+0%
Overnight Fund	sarimax_cal	q75	7	9,395	43.1	-34%	292,451	-37%
Overnight Fund	sarimax_cal	q95	3	27,588	51.1	-21%	24,119	-95%
Overnight Fund	lgbm_q_oosq	q75	7	14,948	35.4	-45%	146,666	-69%
Overnight Fund	lgbm_q_oosq	q95	2	21,788	41.9	-35%	75,850	-84%
Overnight Fund	snaive	q75	4	16,809	37.6	-42%	162,508	-65%
Overnight Fund	snaive	q95	2	31,578	58.9	-9%	39,496	-92%
Overnight Fund	ets_ratio	q75	6	11,104	42.1	-35%	279,762	-40%
Overnight Fund	ets_ratio	q95	2	29,060	54.6	-16%	40,461	-91%

Does q75 vs q95 flip?

Category	Model	(a) q75	(a) q95	(a) cheaper	(b) q75	(b) q95	(b) cheaper	Flips
Liquid Fund	ets_ratio	66.4	124.0	q75	289,238	19,604	q95	**yes**
Liquid Fund	lgbm_q	79.0	134.4	q75	536,914	149,613	q95	**yes**
Liquid Fund	lgbm_q_oosq	94.2	142.9	q75	456,274	122,700	q95	**yes**
Liquid Fund	sarimax_cal	50.0	114.3	q75	255,233	114	q95	**yes**
Liquid Fund	sarimax_cal_oosq	53.3	85.3	q75	284,587	38,668	q95	**yes**

Category	Model	(a) q75	(a) q95	(a) cheaper	(b) q75	(b) q95	(b) cheaper	Flips
Liquid Fund	snaive	56.7	168.0	q75	136,712	168	q95	**yes**
Overnight Fund	ets_ratio	42.1	54.6	q75	279,762	40,461	q95	**yes**
Overnight Fund	lgbm_q	50.9	48.9	q95	386,947	231,981	q95	no
Overnight Fund	lgbm_q_oosq	35.4	41.9	q75	146,666	75,850	q95	**yes**
Overnight Fund	sarimax_cal	43.1	51.1	q75	292,451	24,119	q95	**yes**
Overnight Fund	sarimax_cal_oosq	40.8	47.6	q75	171,635	56,810	q95	**yes**
Overnight Fund	snaive	37.6	58.9	q75	162,508	39,496	q95	**yes**



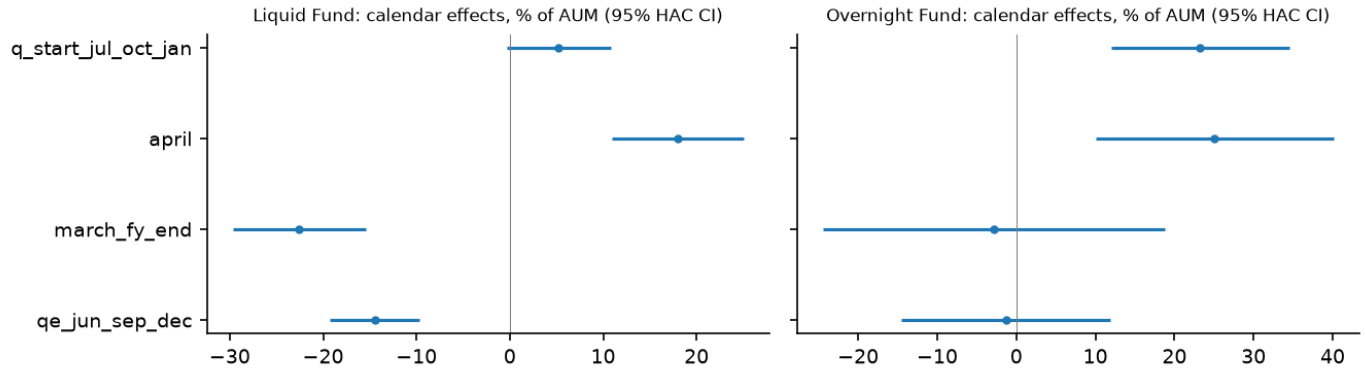
Yes, the conclusion flips. Under reading (a) q75 is cheaper than q95 in 11 of 12 category × model cases and q95 in 1; under reading (b) q75 is cheaper than q95 in 0 of 12 category × model cases and q95 in 12. The cheaper policy changes in 11 of 12 cases.

Why (a) is primary.

1. The brief puts the penalty next to "yield given up on idle cash", which is a rate. The natural reading is a 3× rate on the rupees that are short, so both terms are in the same unit. That gives a well-posed newsvendor problem whose optimal buffer is the 3/(3+1) = 0.75 quantile; `policy_quantiles` in `config.yaml` sets it, and it was not tuned on the test window.
2. Under (b) a rupee short costs ₹3, while a rupee idle costs s/1200 ≈ ₹0.00008 a month at the 0.10% floor, a ratio of about 36,000:1. The optimal quantile is then about 0.99997. Any buffer below the largest plausible outflow is dominated, the spread data play no role, and the stored 2.5–97.5% quantiles cannot even reach the optimum. A 300% charge on principal is also not what a shortfall costs a liquid fund, which can meet redemptions by selling T-bills/CPs or by short-term borrowing (SEBI (Mutual Funds) Regulations 1996, Reg. 44(2), allows borrowing up to 20% of net assets for up to 6 months for redemptions).

What to do with it. If the judges or the fund intend (b), use q95: for November 2026 that is the "Buffer q95" column in §5. Under (a), q75 is the cost-minimising rule. The q75 buffer from `sarimax_cal` beats the baseline under both readings.

4. Drivers



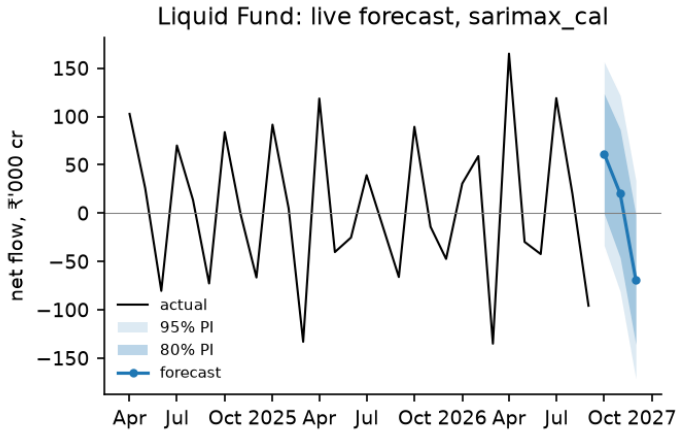
- **In-sample HAC regression** (`outputs/driver_regression.csv`, flow as % of prior AUM):
  - Liquid: Jun/Sep/Dec advance-tax quarter-ends −14.5 pp, March FY-end −22.6 pp, April +18.0 pp (all $p < 0.001$), R^2 = 0.63 with the calendar alone.

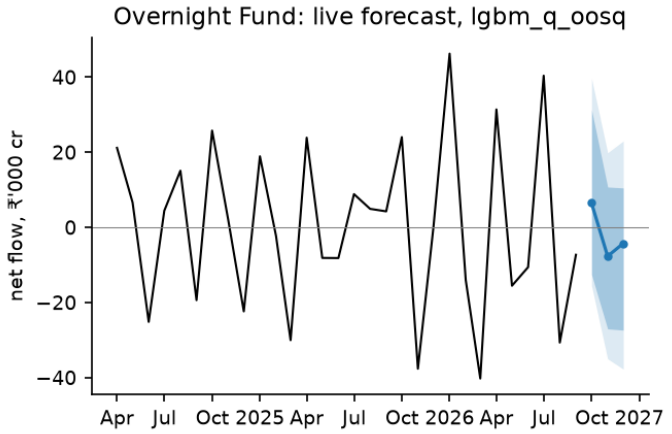
- Adding contemporaneous WACR, spread, net LAF and Nifty raises Liquid R^2 only to 0.63–0.65.
- Overnight: April (+25 pp) and quarter starts (+23 pp) matter. WACR (−3.7 pp per 1 pp, p = 0.01) and net LAF (+3.1 pp per ₹1 lakh cr injection, p = 0.02) are significant contemporaneously, R^2 = 0.40.

- **Out-of-sample** (LGBM feature-group ablation, `reports/tables_ablation.md`): removing the rates, liquidity or equity features does **not** worsen the forecasts. The calendar-only LGBM has the lowest pinball for both categories, so market drivers measured at the origin add no forecasting value beyond calendar effects and target lags at h = 1–3.
- **Regime-break robustness (auditor item 6):** WACR (RBI) is used, not FBIL MIBOR, so the 4-Aug-2025 MIBOR methodology change does not enter the features directly.
  - Break dummies for the SDF floor (8-Apr-2022) and post-Aug-2025 were added to the regression. Calendar coefficients are stable to within 1–3 pp.
  - Post-Aug-2025 is +5.1 pp for Liquid (p = 0.03), which is a level shift rather than a rate effect.

5. Live forecast: origin Sep-2026 (as-of 13-Oct-2026), next release = Oct-2026 data (~10-Nov-2026)

Category	Model	Month	h	Point ₹cr	80% PI	95% PI	Buffer q75	Buffer q95	Baseline buffer
Liquid Fund	sarimax_cal	2026-10	1	61,681	[-361, 123,722]	[-33,203, 156,565]	0	17,948	0
Liquid Fund	sarimax_cal	2026-11	2	20,193	[-46,099, 86,486]	[-81,192, 121,579]	14,697	64,892	14,051
Liquid Fund	sarimax_cal	2026-12	3	-68,770	[-135,644, -1,897]	[-171,044, 33,503]	103,966	154,601	47,308
Overnight Fund	lgbm_q_oosq	2026-10	1	6,510	[-12,496, 31,306]	[-15,434, 39,628]	5,110	13,037	0
Overnight Fund	lgbm_q_oosq	2026-11	2	-7,668	[-26,991, 10,767]	[-34,974, 19,881]	21,627	29,972	37,625
Overnight Fund	lgbm_q_oosq	2026-12	3	-4,250	[-27,322, 10,526]	[-37,705, 23,012]	17,450	32,132	0





All models' live forecasts: `reports/tables_live.md`.

Recommended buffer for November 2026 (decision on 1-Nov using the h=2 forecast, q75 policy under penalty reading (a); q95 if reading (b) applies):

- **Liquid:** ₹14,697 cr from `sarimax_cal`. The q95 buffer is ₹64,892 cr; the baseline is ₹14,051 cr.
- **Overnight:** ₹21,627 cr from `lgbm_q_oosq`, the nested-selection pick. The q95 buffer is ₹29,972 cr; `sarimax_cal` would give ₹14,745 cr; the baseline is ₹37,625 cr.

October 2026 point forecast (one figure per category). Rule, stated in `config.yaml` (`submission: rule: nested_selection, horizon: 1`) on 29-Oct-2026, before AMFI publishes October data (~10-Nov): each category's figure is the h=1 forecast of the model chosen by nested selection, the same model that sets the buffer.

- **Submitted October 2026 forecast, Liquid Fund:** +61,681 cr (`sarimax_cal`, h=1; 80% PI -361 to 123,722; 95% PI -33,203 to 156,565)
- **Submitted October 2026 forecast, Overnight Fund:** +6,510 cr (`lgbm_q_oosq`, h=1; 80% PI -12,496 to 31,306; 95% PI -15,434 to 39,628)

Why `lgbm_q_oosq` and not `sarimax_cal` for Overnight. Nested selection picked `lgbm_q_oosq` on Oct-2024 to Sep-2025. On the later evaluation window it ranks 9 of 10 and `sarimax_cal` ranks 3 of 10. Switching to `sarimax_cal` because it did better on the evaluation window would mean choosing on the window that is meant to stay untouched, which is what nested selection exists to prevent. v0.2 quoted `sarimax_cal` for both categories without a stated rule; that is withdrawn. For reference only, `sarimax_cal` gives +18,003 cr for Overnight October (80% PI -4,490 to 40,496).

December, an advance-tax month, is expected to see a large Liquid outflow: -68,770 cr, with a q75 buffer of ₹103,966 cr.

6. Data gaps and risks

- AMFI data are first releases, and 16 release dates are imputed (month-end + 13 d).
- The daily RBI series before 17-Sep-2026 come from a third-party DBIE mirror, verified on one day. 5 days are missing in Sep-2026. Net LAF is unavailable before Jul-2020 (LightGBM treats it as missing).
- The evaluation covers only 24 targets; coverage CIs are wide and the forecasts overlap.
- Nested selection on 12 months is unstable (see Overnight).
- The cost function uses an ex-post realised monthly spread that is floor-dominated.
- Category-level net flow is not a single fund's gross redemptions.
- The MIBOR/SDF breaks are tested only in-sample.

7. Auditor findings

#	Finding	Verified?	Fix	Regression test	Effect
1	Apr-18 to Mar-19 files have only a combined Liquid/MM row	Yes: old sponsor-wise layout	Panel starts 2019-04 (`config.data.panel_start`); the parser marks that layout and returns no rows	`test_pre_2019_files_yield_no_category_rows`, `test_panel_starts_2019_04_with_90_months_each`	90×2 panel, no category mixing
2	42 .xls files are 404 HTML	Yes (download log)	Files deleted; loader raises `NotAWorkbook` on non-workbooks and logs `REJECTED`	`test_html_404_rejected_and_logged`	102 real files, 0 rejected now on disk
3	xlsx named .xls; identity check	Yes: 75 ZIP, 27 OLE2	Engine chosen by magic bytes; hard assert that sales−red−net is within 0.1	`test_engine_from_magic_bytes_not_extension`, `test_flow_identity_within_tolerance`, `test_matches_independent_auditor_extract`	Max residual ~1e-9; all 180 rows match the auditor
4	Release lag 6–13 d	Yes: observed 8–13 d	Origin as_of = release stamp or month-end+13; drivers need `available_date` ≤ as_of; OOS-PI only uses released targets	`test_info_set_*`, `test_driver_not_available_by_release_is_dropped`, `test_forecast_invariant_to_future_data[*]`, `test_ml_features_equal_under_truncation`, `test_backtest_rows_are_out_of_sample`, `test_oos_calibration_only_uses_released_errors`, `test_release_dates_after_month_end`	Backtest is leakage-free by construction; mutation check confirms the tests bite
5	Report MAE, sMAPE/MASE, median APE	Agreed	`metrics.summarize` adds median APE, sMAPE, WAPE, MAE, RMSE, MASE, scaled pinball, MAPE excluding small months, and Wilson CIs	`test_point_metrics`, `test_coverage_and_wilson`, `test_pinball_sign_and_values`	snaive reproduces the auditor's numbers exactly
6	MIBOR methodology break 4-Aug-2025	Yes, but MIBOR is not used (WACR is)	Break dummies (post-Aug-2025, post-SDF Apr-2022) in the driver regression; LGBM rate-feature ablation	(driver regression output)	Calendar effects stable; rates add no OOS value

Other auditor points:

- No tuning on the test window: hyper-parameters are fixed and selection is nested.
- Horizon-specific out-of-sample PIs: the `_oosq` variants.
- Wilson CIs on coverage: added.
- Provisional-data note: added.
- The DBIE mirror is flagged as third-party.
- SDF break: dummy added.
- Dashboard numbers trace to CSVs: each block names its source file.
- The baseline uses actual(t−12): `test_policy_and_smly_baseline_use_actual_t_minus_12`.

8. Next steps

1. Score the October 2026 figures in §5 when AMFI publishes it (~10-Nov), and extend the evaluation with gross redemptions (a fund-level buffer needs gross, not net, flows).
2. Use a longer selection window (or forecast combination of `sarimax_cal` + `snaive_ratio`) in place of 12-month nested selection, which proved unstable for Overnight. Also add day-of-month intramonth timing of advance-tax outflows (around the 15th) for intra-month buffers.
3. Replace the third-party DBIE mirror with direct RBI DBIE downloads, and fill the 5 missing September days from RBI weekly statistical supplements.

Local application

Everything above can be reproduced with `python -m liqfc run`. The interactive local dashboard (`python -m liqfc app`, http://127.0.0.1:8501) reads only `outputs/`. Its τ, horizon, spread and penalty controls recompute the decision costs from the backtest quantiles, actual flows and realised spreads. A sidebar switch selects penalty reading (a) or (b), and the Decision-cost tab shows both readings side by side. At τ = 0.75/0.95 with the default settings it reproduces `outputs/decision_costs.csv` exactly under both readings (tests/test_app.py). `python -m liqfc refresh` pulls new months from AMFI, NSE, FBIL and RBI. See README.